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EEEC216/214

COMPUTER TROUBLE
SHOOTING NOTE
NOT FOR SALE

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FAULT DETERMINATION AND REPORT

THE PROCESS OF FAULT DIAGNOSIS

The universal troubleshooting process can be broken into four basic steps thus;

STEP1 DEFINATION of SYMTOMS

This involves knowing what the system or equipment is supposed to do when it is operating normally. In addition you must be able to determine or recognize when the normal job is not done. The normal and abnormal system condition is then analyzed in other to raise the question where is the fault emanating from? And what is the cause? it is essential to note down as many symptoms as possible as this would help in refreshing ones memory when troubleshooting in subsquence events of failure.

STEP2 IDENTIFIED AND ISOLATING POTENTIAL SOURCE OF PROBLEM

Tis involve localizing the problem to a functional unit of the system or equipment. It is essential to be sure of the source of the problem. At this stage one need to know if the fault is of software or hardware or even a system sub-assembly as a whole. if the source of the failure is found, such piece of hardware can be isolated and check for problem areas by using approved diagnostic tools or a swap of the suspected sub-assembly or unit.

STEP3 REPLACEMENT / CORRECTION

Having trace the source of failure to the units or sub=assembly level the next thing to do is to troubleshoot to the Circuits / components in the fault unit or sub assembly;

However, in computers and most of their peripherals, it is easier to replace a sub-assembly outright, rather than trying to troubleshoot to circuit components level. It is highly unlikely that your knowledge of replacement of circuits component with ease. Nowadays, manufactures are in the habit of producing in large numbers sub-assembly or units of component system parts. Those parts can be replaced easily with new functional ones the manufacture's part number for the

units or sub-assembly is known. If you need to suspend troubleshooting as you want, the arrival of ordered parts, it is adjustable to collect all the component of the sub-assembly or the faulty units gathered them in to a good quality anti-static bags and label them clearly before storing them

Step 4 retesting the units

This involved checking the unit or sub-assembly so replace or repaired to ascertain a normal working condition of the unit or sub-assembly

1.2 FAULT REPORTFORM

Fault reporting is a maintenance concept that increases operational availability and that reduced operating cost through three criterions

- Reduced labor; intensive diagnostic evaluation.
- ✗ • Eliminating diagnostic testing down time
- Provide notification to user for degraded operation

Although the format for reporting fault may vary from one person to another depending on the purpose a computer is serving, fault report form generally seek to record failures and to keep up-to date events or activities of the user with respect to the system as a whole or its sub-assembly. A computer system fault report form usually contained the following items

- i. The type of system or sub-system assembly
- ii. The model of the system
- iii. Software and hardware details
- iv. Work station ID [for interconnection system].
- v. Description of the fault hardware or software with data, etc

Example of fault report form and their role

- a) Pc configuration form ; it is used when installing and configuring the PC and its expansion board
- b) BIOS Upgrade form ; It is used when planning a BIOS upgrade
- c) System CMOS sheet ; it is used to record parameters of the SET-UP menu

Understanding power supply problems

A power supply pack is a compartment inside the system units that provide a variety of direct current DC or voltage which are required to energized the motherboard and drives.

The input voltage to the power supply pack is alternating voltage. this voltage is normally step-down then rectified in to alternating ripples and with the help of appropriate filtering circuit, it eventually become direct voltage DC the voltage dividing network component is responsible for splitting the direct current into various level and different polarity

The main characteristic of good power supply is its ability to withstand a great deal of electrical stress arising from the conversion of AC to DC. The electrical stress could be inform of heat spikes voltage surge etc.

The vast majority of consumers PC user standard ATX power supplies smaller units and units specially design for enterprise and servo applications are also available for common desktop system ATX power supplies as it.

When searching for a power supply it could be on three crucial features; power output rails and efficiency. Other specification and features important too but those three directly affect the power supply units (PSU)'s performance

OUTPUT POWER Manufactures usually List their power supply output in watts. A higher watt PSU can supply more power. Desktop power supplies have a power output rating of from 200watts to 800watts.

Ideally a PSU will deliver plenty of power to your components and offers some extra head room in case you want to attach additional components later. Most PSU hit their peak efficiency levels with load as range of 40-8%. Building to about 50-60% of the PSU capacity is adjustable to achieve maximum efficiency and yet leave room for future expansion. For instance, if the maximum power or total design power of your system presents components is 300watts, a 600watts PSU will be good for it. In a higher end system loaded watts components at 700watts, 1200watts PSU would be a good fit you can get by with lower capacity units if you

don't think you will ever need to expand your system, and then go for higher capacity PSU

EFFECIENCY OF PSU

A power supply efficiency is important because higher – efficiency units tend to have better component west less power and generate less heat all of which contribute to less fan noise. A power system with an efficient rating of 80% produce 80% Of its rated wattage as power to your system while losing the 20% as heat

THE TYPES OF RAILS

In addition to identifying output power, manufactures, specify the number of +12V rails their PSU contains a single rails power supply has a single high power +12v rails for feeding power to hungry system components. A multi rails units divides its power output between two or more +12v.

In a single rail design, all the power from the supply will be available to any component computer to unit regardless of the connectors or cables used. In the event of a failure, however, a single rail power supplies has the potential to shoot much more current in to your components

Meanwhile, the main advantage of multi rail PSU is that it can't share power among the different rails for example. If you connect 30 amp worth of components to a +12v rail with a 25 amp maximum rating the mismatch will trigger an over current protection mechanism and shutdown even though other rails may be available with plenty power to spark consequently, with a multi rail PSU you may pay attention to which components you have plugged in to which rails unlike with a single rail power supply.

On the other hands, that disadvantage becomes a major advantage if you ever encounter a catastrophic failure. The over current protection mechanism in a multi rail power supply monitor each rail will shoot the whole units down if the detected an overload on any of the rails. The over current protection (OCP) on single rails units

From a performance standpoint, both PSU works equally well, but bottom line is, if you're building especially powerful system and rails OCP provide an extra layer of safety in case something short circuit lessening the odds of frying your costly components during a computing catastrophic.

EFFECTS OF INADQUATE POWE SUPPLY

Inadequate power supply to the system could cause the following failures

- a) Unpredictable power lockup
- b) Random memory failure
- c) Disk access problem
- d) Corruption of file

PROTECTING THE PSU

1. Ensure that a voltage stabilizer is connected to the PSU in place where the supply mains is not table
2. Always check the power rating of the PSU before subjecting it to loading
3. Note the efficiency of the power supply unit so as not to over load the PSU
4. If you suspect under performance, test the voltage rating for the various conductors where there is variation as compared to the standard values, the PSU be replace outright with another one with at test 50watts pin power rating more than the faulty ones but of a similar rails.
5. In the event of long usage of power system, the PSU should be subjected to cooling by placig the system unit in a well-ventilated space.

MEMORY FAILURE AND TROUBLESHOOING

Memory is the corner store of modem computing the overall computer system performance can largely be attributed to the interaction between the memory and the central processing unit (CPU).

FUNCTION OF THE MEMORY

The memory basically perform two function

- Storing of programmed codes
- Storing of data to be processed by the CPU

TYPES OF MEMORY

The type of memory include

DRAM, VRAM, EDO, SDRAM, DDRSDRAM, etc

MEMORY PACKAGE TYPES AND STRUCTURES

There are four package styles for memory devices

- a) Dual in-line package; this memory styles uses through-mounting technique and it is compatible with IC sockets
- b) Single in-line package ; this styles has use solely by Intel 286 and 386 model has since become excellent due to demand for large space
- c) Small outline "J" lead package; this styles also uses surface mounting technique. The protruding leads are bent in form of letter "J" under the package.
- d) Thin, small outline package; it uses surface-mounting package, normally in a smaller package for narrow-space. It is used by notebook system device.

ADD-ON MEMORY DEVICES

There are three basic types, they are,

Proprietary add-on modules

Single inline memory modules (SIMM)

Dual inline memory modules (DIMM)

The SIMM and DIMM are smaller in size, light weight and they contain a relatively large block of memory as compared to the proprietary add-on modules. Again SIMM and DIMM are standardized, which make them flexible for use to any other computer system, but that is not for the proprietary modules.

CAUSES OF MEMORY FAILURE

The causes of memory failure includes,

- i. using the wrong type of memory for certain devices.
- ii. Incorrect installation /configuration
- iii. Wrong insertion of the pack inside a modules
- iv. Using unsupported modules
- v. Virus attack
- vi. Mechanical failure of the memory itself

TROUBLESHOOTING MEMORY (REMEDING PROCEDURE)

- i. Check for general problems common with new system
- ii. Check the positioning of the modules to be sure it is firmly and correctly inserted
- iii. Check for lose connection within the system
- iv. Ensure that the modules sizes is supported because some module are not compatible with some other machines, when in doubt consult the motherbord
- v. Ensure the uses of correct socket be carefully insert the modules in the right order of the memory banks
- vi. Check the technology of the memory to see if it require jumper to be set when adding memory. Where jumper are required, they must be set correctly otherwise, the system will not detect the memory
- vii. The memory modules could also have problem as a result of manufacturing error or mechanical strains, always test before purchasing a new one

MAGNETIC STORAGE DISK DRIVES PROBLEMS AND THROUBLESHOOTING

The disk and its drives are basically the floppy disk drive and the hard disk drives

FLOPPY DISK

A floppy disk is a storage device that is composed of a ring of thin flexible (i.e floppy) magnetic storage medium encased in a square or rectangular plastic wallet. Floppy disk are red and written by a floppy disk drive or FDD. They have the advantage of being removable. They are normally available in different form

factor or format ranging from 8" to 5.2" and 3.5" and also with different densities such as double density DD quad density QD higher density ED higher capacity floppy disk etc. floppy disk and drive are rapidly phasing out due to their small capacity in term of data storage

MEDIA

A media is a physical material that actually hold recorded information . in floppy disk, the media is a small disk coated on both sides in a precise formulated magnetic material often referred to as oxide layer

COERCIVITY

This is define as the strength with which magnetic particles resist the reversal of polarity in a media. High coercivity is desirable (to small extent) because it give rise to much better signals against background noise, also signal will resist natural degradation due to age, temperature and random magnetic influence. Higher coercivity ensure greater information density for media, the greater the resistive strength of each media particles, the higher the number of bits that can be packed in to smaller area. The reason why smaller HDD have higher storage capacity than the larger ones is its coercivity.

SOME COMMON FAILURE, SYMPTOMS AND TROUBLESHOOTING

1. The FDD rotates a devices but will not seek to the desired tract

Possible cause; this suggest that a head position stepping motor is inhibited or defective but other FDD function well

TOUBLESHOOTING TIP

- i. Begin by dis assembling the computer and removed the FDD,
- ii. Carefully impact the head position assembly and be sure there is no abstraction or broken parts. Examine the mechanical system by installing the disk to be sure that the faults are not of disk alignment gently removing any obstruction you might found with great caution.

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- iii. Apply power to the computer and measure device voltage. if other voltage is low or absent, troubleshoot the power supply
- iv. If the power is okay check the FDD PC board or FDD controller IC on the mother board

Used logic probe to measure the step signal in physical interface (pin 20) when device access is requested there should be a pulse signal as the signal floppy controller attempt to position the read / write head if step pulse are not present the floppy drive controller is probably defective and should be replaced. If step pulse is present at the interface check the pulse in to the coil driver circuit. An absent of pulse into the coil driver circuit indicate a faulty control circuit IC. If there is pulse at the coil driver measure the pulse to the stepping motor. If no pulse leaves the coil driver, replace the coil driver IC. If pulse are good at the stepping motor but motion is taking place replace the stepping motor. Or drive PC board or the entire drive outright.

THE FDD HEAD SECK PROPERLY, BUT THE BUT THE SPINDLE DOES NOT TURN

Possible causes; This fault suggest that the spindle motor is inhibited or defective but all other function are working properly.

Troubleshooting tips

- i. Remove all power from the computer, disassemble and remove the FDD
- ii. Carefully inspect the spindle motor drive belt and spindle assemblies. Be sure that there is no obstruction and ensure that the belt is reasonably tight.
- iii. Examine the FDD with a diskette inserted to rule out disk insertion or alignment problem. Try many different diskette.
- iv. Check the voltage levels
- v. If the FDD is mechanically sound and appropriate power is available, check the PC board or the FDD controller board.
- vi. Using logic probe measure the motor on signal at pin 16 when drive access is requested. The signal should be true (active low) but if the signal missing then the floppy drive-controller board is probably defective and should be

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replaced. If the motor on signal is present at the interface, check the signal driving the servo-circuit. A missing motor on signal at the servo circuit suggest a faulty control – circuit IC. If the signal reaches the servo circuit, the servo IC is probably defective and should be replaced, but it should be better to replace the spindle motor and PC board assembly as a unit or even the FDD outright should be replaced.

- vii. When data can be writing to the floppy drive, used logic probe to measure the gate and wright data line. During the wright cycles the wright gate should be logic 0, and should read a pulse signal as data flows from the floppy controller IC to the driver. If the two wright signals appears as expected, check the analogue signal to the R/W heads with oscilloscope. If there is no analogue write signal, replace the defective control-circuit IC. If analogue signal are present to the heads try replacing the heads or the entire head carriage assembly or the entire drive outright.

THE HARD DISK

Hard disk (HDD) is one of the core component of any computer. it is a non-volatile storage drive that store digitally encoded data. The files are stored in such a way that they can acquire at any time. It can contain huge amount of data. A HDD is physically composed of a series of flat, magnetically coated platters stacked on a spindle. The spindle turn while the head move between the platters in random radially reading/ writing data on to the platters. HDD form factors ranges between 5. 25 and to 1inch

THE FLOPPY DISK WILL NOT READ /WRIGHT TO THE DISKET WHILE ALL OTHER OPERATIONS APPEARS NORMAL,

Possible cause; this suggest there is no communication between the drive controller IC and the R/W head assembly.

- i. **Troubleshoot tips;** begin by trying a known-good, properly formulated diskette in the drive. If a known-good diskette does not solve the problem, trying clearing the R/W head but not run the drive with a head cleaning disk inserted for more than 30second at a timeto avoid damage of the heads by fiction'
- ii. Determine whether the trouble is in the FDD assembly or the floppy – controller IC
- iii. If data cannot be read from the FDD use logic probe to measure the read data signal when the disk is idle, the read data line should be constant logic1 or logic 0. During the read cycles these should be pulse signal as data move from the drive to the floppy controller board. If no pulse appears at the read data line during a read cycles use oscilloscope to measure analogue smaller from the R/W reads. If there is no signal from the R/W heads, the controller circuit IC is probably defective and should be replaced.
- iv. If you are un able to replace the IC, you can replaced the drives is control PC board, or the entire drive outright, but if there is a pulse signal during read cycles, the floppy disk controller board is is probably defective.

HDD OPERATION

When the drive is powered the motor begin to spin the spired that holds the magnetically coated substrate material in turns the platters are spin at a corresponding speed.to that of the spindle. Magnetic R/W heads which are anchored by the actuator arm are brought in to close proximity to the spinning platters to detect or create flux transition as required.

TYPES OF PLATTER

There are two types of platter this include

- Metallic alloy (e g magnesium aluminum alloy)

- Glass or glass ceramic composite (e.g. metal). The glass or glass ceramic composite are more rigid than metallic alloy platters more so the glass platters are more thermally stable than the alloy platters

FUNCTION OF HDD

- Holds the OS
- Store large volume application and files.
- Provide virtual memory

POWER - MODE DEFINITION

The need for power conservation in the computer system overall power management has given rise to several levels of power demand on a host system.

* HDD is typical operation line in any of the following power mode.

1. SPIN-UP: ^{Wake up} This occurs at initial application of power in watts is required for this mode
2. Seek: ^{seek} This mode prepares the device to locate the required track for reading or writing it requires between 8.5 to 9 watts
3. Read/write: This mode occurs at the instance where data is being read/written. It requires 5W
4. Idle: This is when the activity of the actuator head is suspended. It is a basic power conservation mode that requires just 4W
5. Stand by: At this mode, the spindle motor has stopped moving. This is the main power conservation mode which requires just 1W

TYPES OF HDD

The HDD are usually identified by their connection are;

- IDE/DAPA; This has 40pin ribbon connecting that can connect up to two device
- SATA; This has 7 conductors and can only connect a single HDD to a single connector on the SATA controller found on the mother board

- iii. SCSI; This has 50-60pin multiple SCSI device can be connected to a single channel.
- iv. USB; This could actually be a PATA or SATA device in single enclosure with a conversion for connection to USB interface

REASON FOR HDD FAILURE

Broadly speaking there are four ways by which failure could occur in a HDD.

- a) Firm ware corruption (e.g Air filter problem)
- b) Electronic fault (e.g damage blocks)
- c) Mechanical fault (e.g head crush)
- d) Logical fault (e.g power supply fault)

HDD FAILURE, SYMPTOMS AND SOLUTION

HDD failure can occur either gradually or catastrophically. It is difficult to diagnose gradual HDD failure as its symptoms such as showing down PC corrupted files e.t.c. may be caused by other issues like viruses. In most cases of HDD failure the data can be partially or totally recovered in severe cases, the data may get completely destroyed

HDD appears to be completely dead: device does not spin, no indicator light, and probably an error message such as "drive not found" appearing.

Possibility: This problem is probably from the drive connection to the power supply.

TROUBLESHOOTING

- i. Check the 4-pin power connection and ensure they are secured
- ii. Check the power levels of pin 4 and pin 1 which output to be +5V and 12V respectively. Replace power supply if the voltage is unusually low or absent
- iii. Check for damage or worn out cables and replace if necessary.
- iv. Check CMOS setup routine to see that all the parameters entered for the drive are correct else post will not recognize.

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- v. If problem persist the HDD itself might be defective. Try a known good HDD. If it is fails to operate, then replace the drive controller board.

The drive seen to be working but the system will not boot from the HDD

* Possibility: The cause of this fault could be from the boot sector failing, the drive failing or DOS/windows file been corrupted

TROUBLESHOOTING TIPS

- i. Check the drive signal interface at both end and at the controller ends. Replace if damage or worn
- ii. Check CMOS for correct parameter of head, cylinders sector per tract landing zone and write pre compensation. also try Auto-detection
- iii. Attempt booting from FDD and try accessing the HDD. If the HDD is accessible, chances are that, the boot files are missing or corrupt
- iv. Try utility like "FDisk MBR" which will rebuild the device master boot record. Do not re partition or reformat a HDD without you backing up the data in it unless if it is absolutely necessary. Repartitioning should only be done if the controller responds but the drive does not. In such instance, try repartitioning the HDD. If the drive still fails then replace. If the controller doesn't respond, replace the HDD controller.

The drive is active but un able to execute read or write instruction

* Possibility: this fault is is possibly due to a bad sector ID or aging

TROUBLESHOOTING

- i. Begin by checking for the structure problems or the device
- ii. Use dis fix or scandisk to search for load sector in the device. If a fail sector contains part of an EXE or COM files, that files has been corrupted and should be replaced or restore from the back up.
- iii. If the problems cannot be isolated, try a low level formatting this will re write sector ID information.
- iv. If drive manufacturers provide a drive preparation tools, you should back up the drive, run the tools FDisk, format and restore the drive.

HDD performance appears to be slowing down over time in virtually all cases.

Possibility: the possible cause of this fault could be computer virus attacking the system or mostly the fragmentation.

TOUBLESHOOTING

- i. Start a PC with clear boost disk and ensure that no driver has being loaded.
- ii. Run anti-virus checker and ensure there are no memory resident or file base viruses.
- iii. If the system is free from virus, then check for fragmentation using utilities like compress with PC tools or DEFFRAG with DOS, to see the % of file fragmentation. If it is more than 10%, run the fragmentation utility after carrying out the following
 - Reboot the system normally, start windows, access the virtual memory control for the version of windows use and shot down virtual memories
 - Exit windows and boot the system check again
 - Restart defragmentation utility and proceed to defragment the disk
 - When completed, reboot the system normally, start windows, access the virtual memory control for the version in use and recreate a permanent swap files to support virtual memory.

The HDD is not accessible and they are light stay ON continuously

Possibility : This fault is possibly due to wrong placement of signal cable while upgrading or replacing a drive system.

TOUBLESHOOTING

- i. Check the signal cable and ensure that both end of the cable are properly installed. Observe the cable for red or blue stripe on the side and ensure that it is positioned to fit into pin 1
- ii. Try a good cable

The error message "no fixed disk present" appears

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Possibility: This error indicate an installation problem or it could be a warning that that the power supply to HDD is problematic

TROUBLESHOOTING

- i. Check power connectors and ensure that the 4 pin power by is intact
- ii. If the drive is powered by connector, make sure that any interim connections are secured.
- iii. Test the voltage ta pin one and pin 4 to see if are okay. If they are not replace the power supply.
- iv. Check the signs connectors to ensure that the cable is secured at both ends. Replace any worn out or damage cable
- v. If problems continued, check CMOS set up and ensure that the parameters are corrected. Also check for hard ware conflict to ensure that no other expansion device in the system use the same IOR or I/O used by the HDD controller if so, change the resource used by the conflicting device.
- vi. If problem still continues try a known good HDD. If this also fail, then replace the HDD controller board.

"171 error" appears as ao error message.

Possibility: This is an indication that HDD has failed post test. It is apost error.

TEOUBLESHOOTING

- i. Check power parameters
- ii. Check CMOS tor parameter drive parameters
- iii. Try auto detecting and or low level formatting

UNEXPECTED COMPUTER CARASHES

- i. **Possibility:** the HDD has incurred too many bad sectors or bock such that the head may not be able to identified the sectors user requires.
- ii. It is also possible that the HDD has been formatted more its limit
- iii. The HDD to old to be in use

Remedy

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- i. Ensure regular update of the system
- ii. Replace the/HDD after 3-5 years of effective usage

FILES GETTING CORRUPTED ON HDD

Possibility:

- i. Unethical shutting down of the device
- ii. Closing a running programmed accidentally
- iii. Power failure and virus attack.
- iv. Use of computer software.

Remedy

- i. Always shut down the system by following the wright procedure
- ii. Avoid installing suspected, software malicious programs in to the HDD
- iii. Clear and remove unwanted program regularly

UNDERSTANDING CDROM/DRIVE FAILURE SYSMTOMS AND TROUBLESHOOTING

Compact disc (CD) was initially produced to replace a "Ging" phonograph technology which utilized magnetic storage concept. Unlike previous recording media the compact disc recorded data in digital form through the use of physical "pits" and "lands" in the disc using optical means

CD-ROM DRIVE

The CD-ROM drives in computer that red the data discs are almost identified to auto cd players. The main change is in the circuitry. The circuitry is modified to provide additional error detection and correction so as to avoid unacceptable missing of data in the file

DC-ROM DRIVE MECHANICAL OPERATION

A CD-ROM drive operate by using laser to reflect light off the bottom of the disc the reflective light is then read by a photo detector. The drives operation follows these steps

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- i. The laser emits a low energy infrared beam toward a reflecting mirror.
- ii. The servo motor, on command from the microprocessor position the beam on to the correct tract on the CD-ROM by moving the reflecting mirror.
- iii. When the beam hits the disk its refracted light is gathered and focus through the fort lens beneath. The platter, bounce off the mirror and sent toward the beam splitter.
- iv. The beam splitter directly the returning laser light toward another focusing lens.
- v. The last lens direct the the light beam to a photo detector that convert the light in to electrical impulse.

COMPACT DISK MEDIA

The CD plate is made of a poly carbonate water also known substrate measuring 12mm in diameter 1.2 mm thick 15mm hole in diameter, in the center. The water base is stamped with a single physical tract in a signal configuration starting from the inside of the disk and spiraling outward. The tract has a pitch of spiral separation of 1.6 microns (1.6×10^{-9} mm). when examining the spiral tract of a cd-rom a microscope, raised bumps called lands can be seen along the tract

DATA (BITS) ENCODING/READING

The laser used to read the disk would pass right through a clear plastic, so the stamped surface is coated with a laser of the metal (Gold or Aluminum) to make it reflective, then the metal is coated with a thin protective layer of a transient material these translucent and gold layer allows the recorded pits and loads to be read back after recording. Compact Disk (CDS) laser disk (LD) and mini disk uses a technique of data encoding known as eight – to – fourteen modulation (EFM) under EFM rules, the data to be store is first broken into eight - bit blocks (bytes) each eight – bit block is translated in a corresponding fourteen – bit code words using a look up table the 14 bit word are chosen such that binary ones are always separated by a minimum of two and maximum of ten binary zeros. This is because bits are encoded with non-return to zero inverted (NRZI) encoding or module – 2

integration so that the binary one is store on the disk as a change from a land to bit to a land, while a binary zero is indicated by no change'

HANDLING ACD

These are precaution measures that could help to prolong the DDS lifespan, these include

- i. Avoid binding of the CD
- ii. Protect the CD from heat
- iii. Avoid scratching on the CD especially circular scratch which could easily wipe out entire segments of data
- iv. Avoid using chemical to clean the CD
- v. Cleaning should be done by holding the disk from the edges and wipe radially but not in a circular ways,
- vi. Use a clean cloth soaked in a bit of fresh isotropy alcohol to clean , do not use water
- vii. Avoid writing on the disk because the ink may diffuse through the top surface and damage the disk

CLEANING AN OPTICAL DRIVE

The first symptoms of a dirty optical drive is that you get read errors on a data CD or DVD degraded sound or video from an audio CD or DVD – video disk, if this happen, it is often because the disk itself is dirty or scratch, so clean the disk or try a different disk before assuming the device is of fault.

Try loading optical device requires little cleaning. They are well well sealed agist dust and all recent drives incorporate a selfcleaning fens mechanism for routine cleaning wipe the external parts of the device occasionally with a damp cloth. Some drive matters recommend using a drive cleaning kit every two days although it is usually done when read error are noticed.

To use those kits, which are available to wet and dry forms, insert the cleaning disk and access the device to spin the cleaning disk for a few second for a